AI-BASED EARLY CHILDHOOD EMOTIONAL INTELLIGENCE MONITORING PLATFORM IN URBAN AREAS

25-26J-088

Project Proposal Report

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Specializing in Data Science

Department of Computer Science and Data Science

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DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date: 27/08/25.

Signature of the co-supervisor: Date: 27/08/25

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ABSTRACT

Fine motor skills are a central part of early childhood development, as they allow children to perform precise movements that are essential for academic readiness and everyday activities. In Sri Lankan preschools, these skills are commonly developed through the use of structured activities including tracing, drawing, and coloring. Fine motor development is directly linked to self-regulation, which is a key aspect of emotional intelligence, because activities like tracing and coloring require focused attention, patience, and impulse control. By assessing children's performance on these activities, the study measures their degree of focus, which is indicative of both motor coordination and self-regulatory capacities. The main aim of this research is to develop a computer vision based models that combines tracing and coloring activities in order to create a holistic, longitudinal measure of fine motor skills and attentional focus. By combining these analyses, the model provides accurate information about fine motor control and self-regulation, thus allowing teachers and parents to determine areas of developmental need and provide targeted interventions. Monitoring performance over several sessions allows classification into predetermined skill levels, while the incorporation of historical data provides more accurate monitoring and measures aimed at improving both fine motor skill and emotional self-regulation.

Keywords: Fine motor skills, Eye–hand coordination, Self-regulation, Emotional Intelligence, Focus level assessment

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1. INTRODUCTION

Emotional intelligence, usually described as the ability to recognize, understand, manage, and use emotions effectively in oneself and others [9]. By developing emotional intelligence, people learn more about themselves and learn to cooperate with those around them [10]. For early childhood emotional intelligence is crucial because lower emotional intelligence impacts clear thinking process and the education of the preschoolers [11].

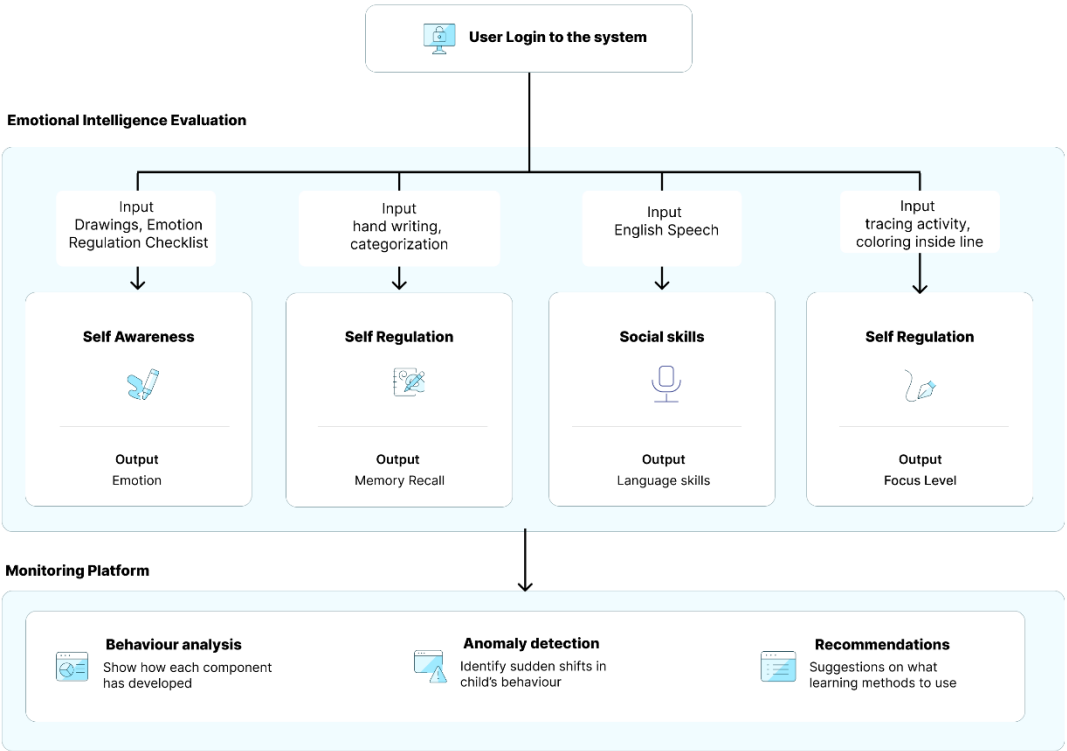


Figure 1:1Overview Diagram

Alongside emotional growth, neurodevelopment and motor development are equally essential in building social and academic skills [1]. Fine motor development, fostered through activities such as cutting, tearing, folding, pleating, gluing, tracing, drawing, coloring, and others supports hand–eye coordination, concentration, and self-regulation. When children are able to control and refine their motor skills, their self-confidence and independence begin to strengthen [2]. Together, these aspects highlight the importance of assessing both emotional and motor development in early childhood education.

1.1. Background & Literature Survey

We conducted an online survey to gather information on how important it is to keep track of emotional intelligence in early childhood. This survey was conducted by sending the online form to parents and the teachers in urban areas in Western province. The survey results will help in building the application in a more successful manner. The form was filled by 64 individuals. Below are the results of the form.

1. What is your role? (ඔබේ භූමිකාව කුමක්ද?)

Parent (දෙමාපියන්)	78
Teacher (ගුරුවරයා)	3
Both (දෙදෙනාම)	6

Figure 1:2Survey Result 1

2. Type of school/setting? (පාසලේ/සැකසුමේ වර්ගය?)

Government (රජයේ)	25
International (ජාත්‍යන්තර)	21
English Medium Pre School (ඉංග්‍රීසි මාධ්‍ය පෙර පාසල්)	12
Sinhala Medium Pre School (සිංහල මාධ්‍ය පෙර පාසල්)	29

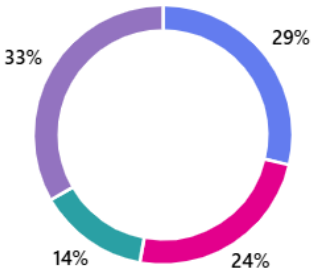


Figure 1:3 Survey Result 2

3. How useful would you find a tool that helps you track Memory recall, Focus, Emotion, Language Skill in your child/students? (ඔබේ දරුවා/සිසුන් තුළ මතකාවර්ජනය, අවධානය, හැඟීම්, කථන කුසලතා නිරීක්ෂණය කිරීමට ඔබට උපකාරී වන මෙවලමක් ඔබට කොතරම් ප්‍රයෝජනවන්ද?) More details

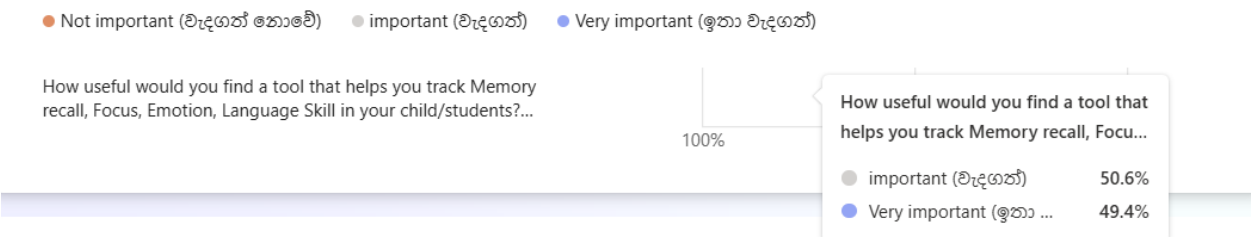


Figure 1:4 Survey Result 3

4. How do you currently monitor Memory recall, Focus, Emotion, Language Skills? (ඔබ දැනට මතකාවර්ජනය, අවධානය, හැඟීම්, භාෂා කුසලතා කුසලතා නිරීක්ෂණය කරන්නේ කෙසේද?) More details

Figure 1:5 Survey Result 4

5. Primary language used at home ? (නිවසේදී භාවිතා කරන ප්‍රාථමික භාෂාව ?) More details

Figure 1:6 Survey Result 5

6. Do you use English as the secondary language? (ඔබ ද්විතියික භාෂාව ලෙස ඉංග්‍රීසි භාවිතා කරනවාද?) More details

Yes	69
No	17

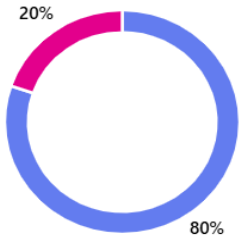


Figure 1:7 Survey Result 6

7. What activities does your child do most? (ඔබේ දරුවා වැඩිපුරම සිදුකරන ක්‍රියාකාරකම් මොනවාද?) More details

Writing (ලිවීම)	38
English speaking (ඉංග්‍රීසි කථා කිරීම)	36
Drawing (ඇඳීම)	73
Tracing	48
Other	10

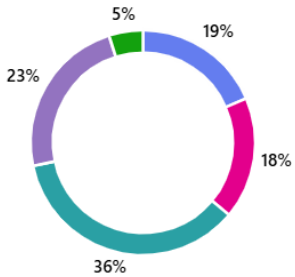


Figure 1:8 Survey Result 7

8. How often does your child write letters at home ? (ඔබේ දරුවා නිවසේදී කොපමණ වාරයක් අකුරු ලියනවාද ?) More details

Mostly English letters (More than 4 days a week) (බොහෝ විට ඉංග්‍රීසි අකුරු (සතියකට දින 4 කට...	12
Mostly Sinhala letters (More than 4 days a week) (බොහෝ විට සිංහල අකුරු (සතියකට දින 4 කට...	11
Both Sinhala & English letters (Less than 4 days a week) (සිංහල සහ ඉංග්‍රීසි අකුරු දෙකම (සතියක...	23
Does not write letters at home (ගෙදරදී අකුරු ලියන්නේ නෑ)	41

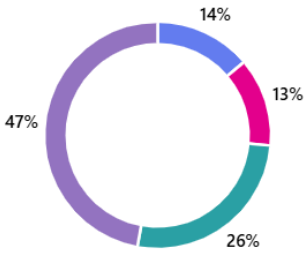


Figure 1:9 Survey Result 8

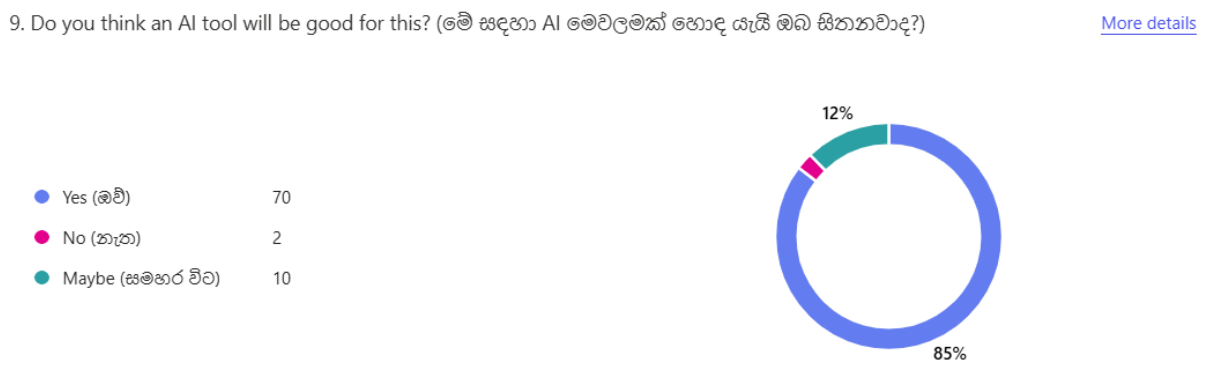


Figure 1:10 Survey Result 9

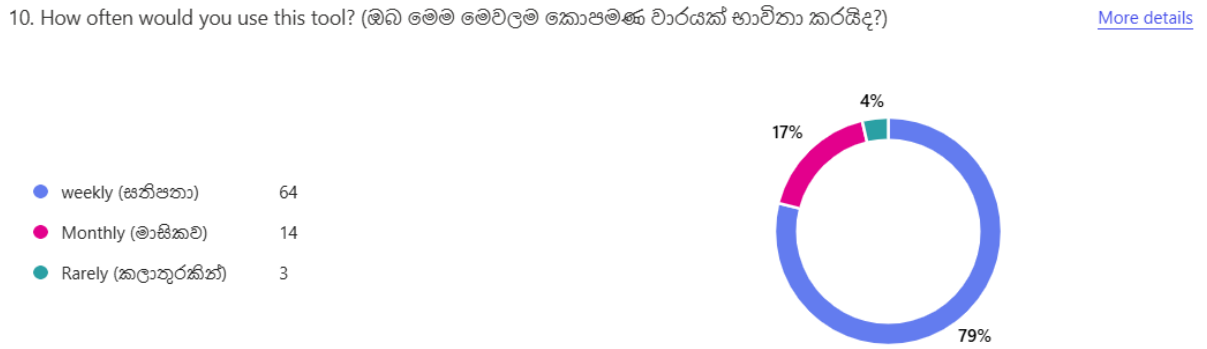


Figure 1:11 Survey Result 10

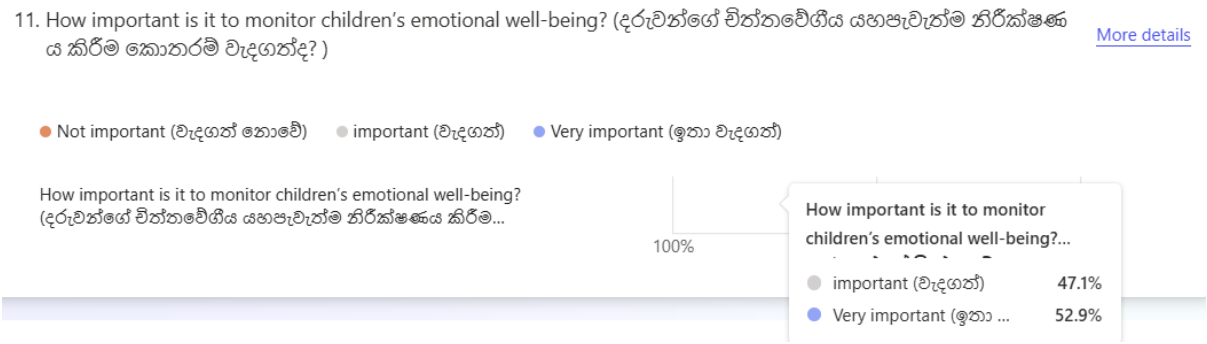


Figure 1:12 Survey Result 11

This survey was filled mostly by parents whose children, go to Sinhala medium preschools, mostly speaking in Sinhala and used English as the secondary language. Most of the responders agreed, a tool that can track memory recall, focus, emotion and language skill will be useful. Currently no formal method is being used for these tasks. 85% of the people agreed that an AI tool would be helpful. Out of 87 people who filled the form, 64 responded that this should be assessed in a weekly basis. Overall, every single respondent agreed that monitoring a child's emotional well being is a crucial tack.

The survey findings revealed that most children in preschool regularly engage in coloring and tracing activities. My study uses coloring and tracing activity images to assess focus level and fine motor development, ensuring validity and avoiding conflicting results from evaluating fine motor control with a single task. Previous research [3] indicates that evaluating fine motor control through a single task may yield inconsistent results; therefore, the employment of multiple tasks guarantees a more reliable assessment.

1.1.1. Coloring and Focus Level

Coloring is the one of the types of activities that children like to increase their creativity and imagination to develop their fine motor skills. And also those activities help to improve the focus level.[4]

One study shows that coloring activities improve children's fine motor skills**.** Regularity (79%), completeness (71%), neatness (67%), hand-eye coordination (81%), and accuracy (86%) all increased, indicating notable development in fine motor abilities[5].

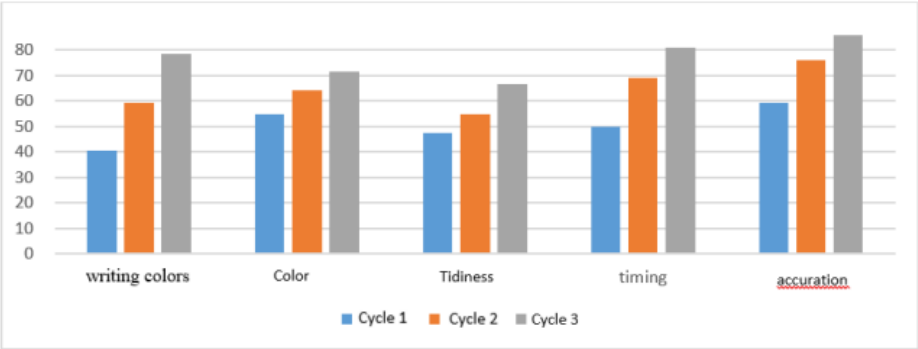


Figure 1:13 Fine motor skills with coloring activities[5]

That study have evaluated the coloring scoring method to assess the fine motor development considering those areas.

The measured aspect	Score 3	Score 2	Score 1
color scribbles	regular	less regularly	not regular
coloring	fill the image field	almost fill	not full
neatness	not out of line	slightly out of line	out of line
hand and eyes coordination	Timing /fast Accurate	Less fast Less precise	Not fast Not precise

Figure 1:14 Grading Criteria For Motoric Fine Through the coloring [5]

1.1.2. Tracing and Focus Level

Tracing is another fundamental activity that supports children's fine motor development by strengthening hand eye coordination, grip control, and precision in movement. Unlike free drawing, tracing is considered an externally cued task where the child follows a given path or boundary. This reliance on external cues requires the child to maintain sustained attention and focus throughout the activity to remain within the lines[3].

The another study further emphasizes that tracing activities enhance hand-eye coordination and accuracy, and children who perform well in tracing tasks tend to show higher levels of concentration and self-regulation[7].

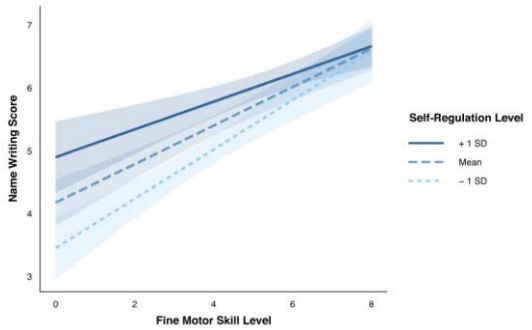
1.1.2. Fine Motor Development and Self Regulation

Research indicates a meaningful statistical relationship exists between children's self-regulatory abilities and their fine motor development. This connection was demonstrated through comparative analysis of writing performance measured at both the start and conclusion of the preschool academic year. The data presentation includes tabular summaries alongside graphical representations for clarity [8].

Variable	Beginning of the Preschool Year	End of the Preschool Year	t	d_{rm} [95% CI]	p
ESI-R Score (Fine Motor Skill)	3.1 ± 2.3	4.5 ± 2.2	12.1	0.61 [0.49 to 0.72]	< 0.001**
HTKS Score (Self-Regulation)	18.4 ± 15.2	29.8 ± 20.9	11.5	0.60 [0.49 to 0.72]	< 0.001**
Name Writing	4.7 ± 2.0	5.6 ± 1.9	7.4	0.43 [0.31 to 0.54]	< 0.001**
Letter Writing	34.8 ± 17.0	40.9 ± 17.3	8.1	0.36 [0.27 to 0.45]	< 0.001**
Word Writing	11.9 ± 9.6	13.4 ± 10.8	2.7	0.15 [0.04 to 0.26]	0.006**
Story Writing	2.8 ± 1.1	3.2 ± 1.2	7.0	0.42 [0.30 to 0.55]	< 0.001**

Note. * denotes significance at $p < 0.05$. ** denotes significance at $p < .001$. These data reflect comparisons of the children ($N = 272$) who were tested at both the beginning and the end of the preschool year.

Figure 1:15 The table shows how children improved in fine motor skills, self-control, and four writing activities from the start to the end of preschool [8]



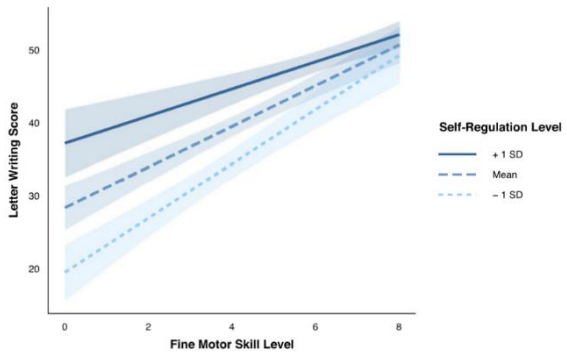


Figure 1:17 The chart displays the strong link between children's behavior control and hand coordination, especially for name writing skills after preschool [8].

Figure 1:16 The chart displays the strong link between children's behavior control and hand coordination, especially for letter writing skills after preschool [8].

Studies suggest that when children develop good hand skills, they become better at controlling their emotions, focusing, and not giving up, which helps them succeed in learning and school.

1.2. Research Gap

Prior research has typically assessed only a single activity either tracing or coloring and has rarely examined how a child's attention. This limitation reduces the accuracy of evaluating children's focus levels and fine motor skills. Therefore, there is a need for a more reliable and comprehensive approach that integrates multiple fine motor activities and analyzes performance patterns to provide deeper insights into children's focus and developmental progress.

Research Paper	Inputs	Fine Motor Development	Focus Level	Tracking Over Time	Automated Scoring Method from Model
[3]	Tracing	✓	×	×	×
[4]	Coloring	✓	✓	×	×
[5]	Coloring	✓	×	×	×
[7]	Tracing	✓	×	×	×
My Study	Tracing + Coloring	✓	✓	✓	✓

Table 1:1 Comparison of prior research works for coloring and tracing activities

1.3.Research Problem

In early childhood, neurodevelopment and motor development are vital for building social and learning skills.[1] Self-confidence, self-confidence will emerge psychologically when children are able to control their motor development[2].When fine motor skills are not properly identified, children face challenges that negatively affect their academic performance, learning pace, concentration, and overall educational growth[1,6].

- Difficulty in performing tasks such as tracing, clipping, filling, and tearing.
- Negative impact on higher education outcomes.
- Delays in the overall learning process.
- Reduced focus, attention, and self-regulation.

To address these challenges, this application helps teachers and parents to identify the fine motor development level and focus level of children. By providing clear insights, it supports early intervention and ensures children receive the guidance they need to improve these essential skills.

2. OBJECTIVES

2.1. Main Objective

Help the parent and teacher monitor the child's emotional intelligence development and find whether any critical situation has occurred.

2.2. Specific Objectives

- Analyze coloring activity performance by assessing boundary control, completeness, and scribbling behavior.
- Evaluate tracing activity accuracy by measuring line-following precision, lines complexity and completeness.
- To combine results from both tracing and coloring activities to determine the overall fine motor skill level.
- To measure the focus level of children through completeness, coloring/tracing within bounds, area colored outside the line, and task completion time.

3. METHODOLOGY

This study focuses on children aged 4–6 years from selected urban preschools in Sri Lanka. Participation will be voluntary and conducted with approval from preschool teachers and administrators. The project aims to develop an AI-based system capable of automatically evaluating fine motor skills and focus levels during coloring and tracing activities.

Activities Data

- Children will participate in structured coloring and tracing activities in their usual classroom environment.
- Each activity will be digitally recorded using cameras or scanners, providing data for computational analysis.

Automatic Scoring of Activities

- The models will generate scores for each child's fine motor skills and focus level based on their coloring and tracing activities.
- Scoring criteria (e.g., completeness, boundary adherence, neatness, and level of focus during the activity) were defined in consultation with qualified preschool teachers to reflect real classroom evaluation standards.

3.1. System Overview

The proposed system is an AI-based platform for monitoring early childhood emotional intelligence, focusing on four main areas: emotion identification, memory recall, focus level identification, and language skill identification.

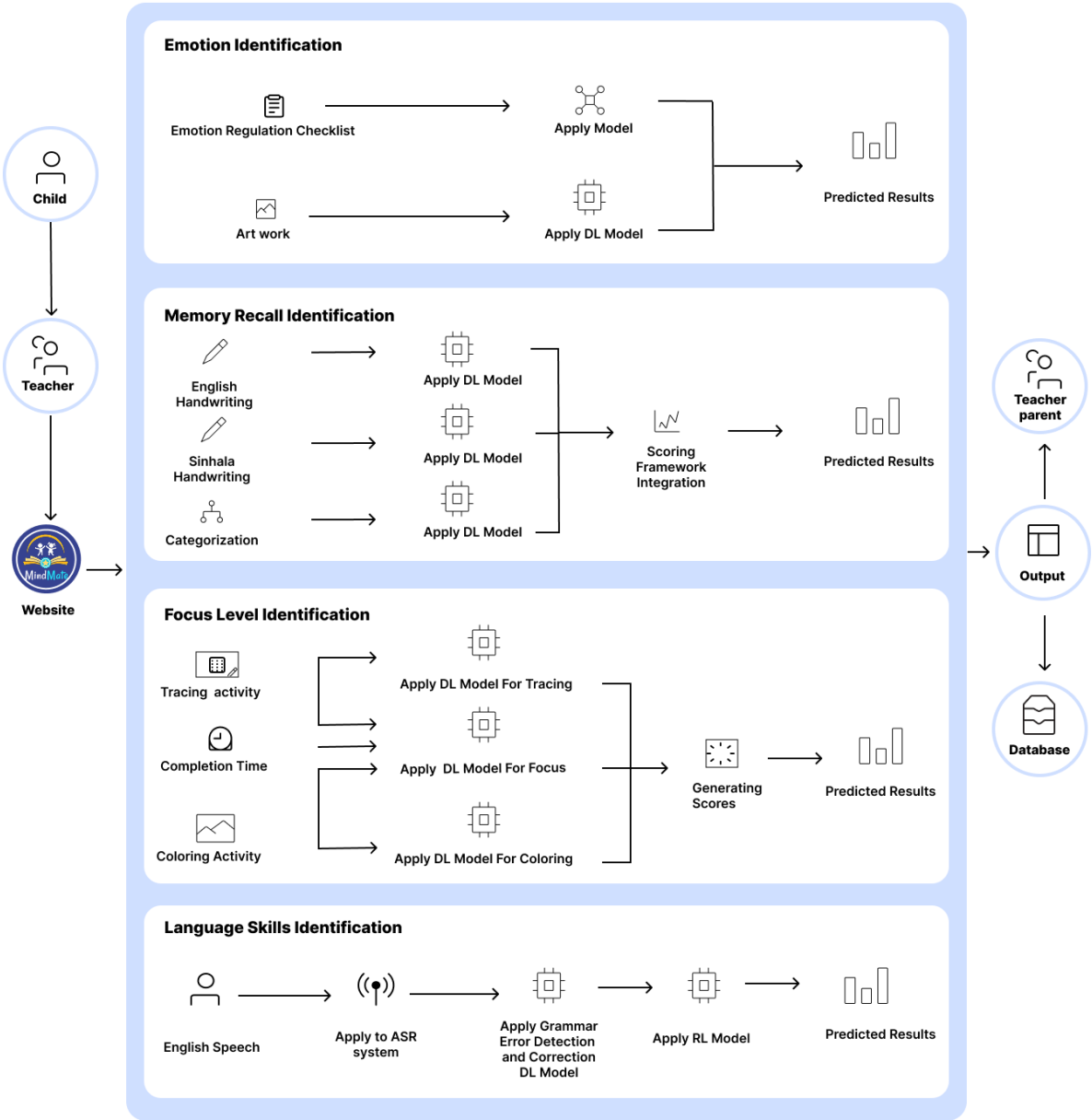


Figure 3:1 System Overview Diagram

My component takes children's coloring and tracing activities as input. The AI scoring models generates scores for fine motor skills and focus, simulating teacher evaluation. Finally, the results are displayed on the dashboard for teachers and parents to monitor the child's development.

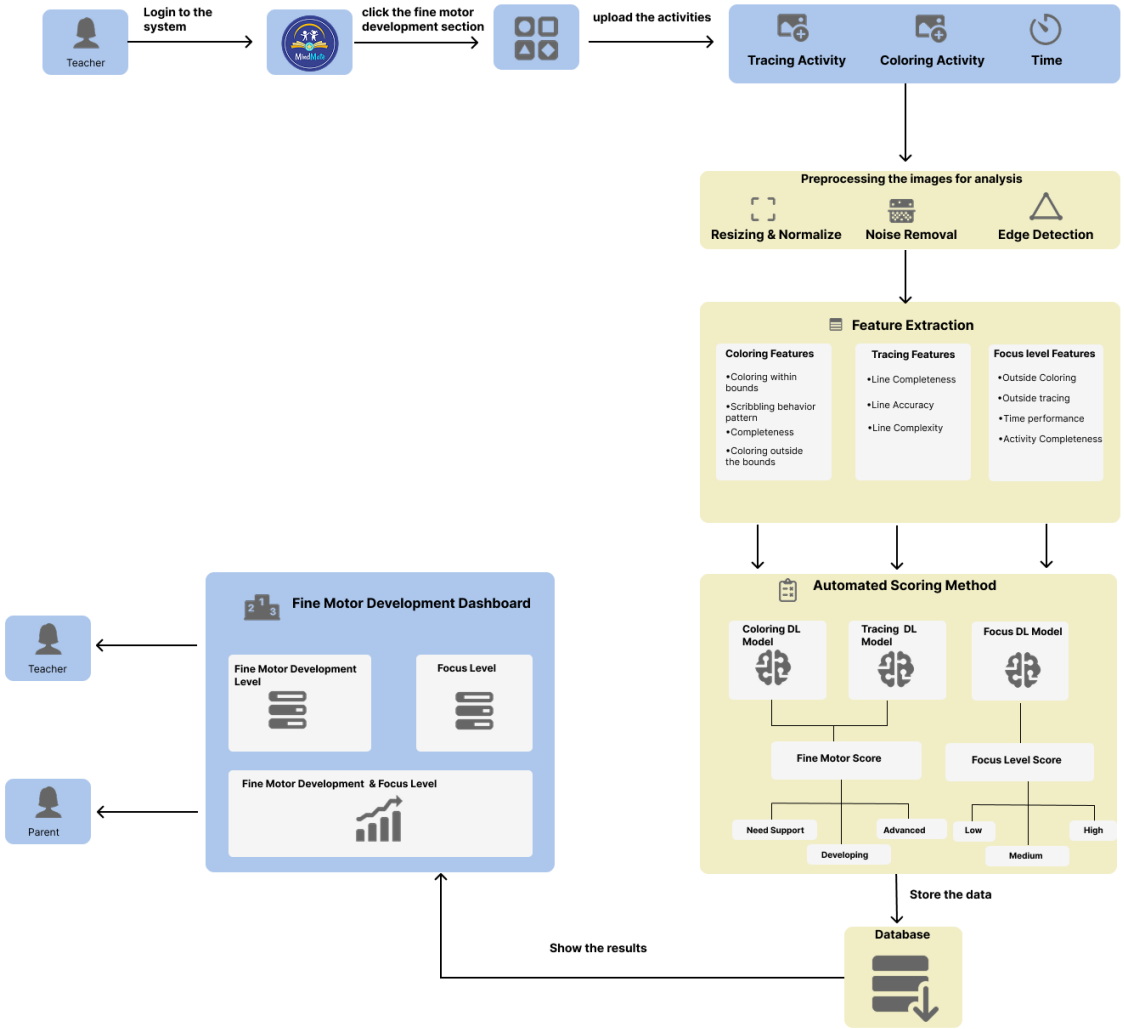


Figure 3:2 System Diagram for Fine motor development and focus

3.2. Tools & Technologies

The system to be built will apply deep learning techniques in automatically assessing the fine motor skills and attentions of children from coloring and tracing. Deep learning is well-suited for this task because it has the capability to identify complex patterns and subtle differences in the images, such as completeness, accuracy, and attentions-related features. Because of the comparatively small dataset, transfer learning will be applied to leverage pre-trained models so that the system learns effectively with small data. Image processing techniques, including image augmentation will also be utilized to augment the size of the dataset and improve model generalizability.

The system will be implemented as a web application for ease of use by teachers and parents. Model training and experimentation will be conducted in Jupyter Notebook environment, while Visual Studio Code will be the main IDE for system development, including both backend and frontend.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

- The system supports uploading children's tracing and coloring activity images.
- The system automatically detect and extract features including completeness, boundary alignment, neatness, and choice of colors.
- The system assess and evaluate the child's focus level during the activity.
- The system should generate scoring metrics for fine motor skills and attention using the identified features.
- The system should store uploaded images, extracted features, and calculated scores securely in the database.
- The platform will offer teachers and parents a simple interface to check student level.

4.2. Non- Functional Requirements

- **Performance**
  Activities uploaded to the system must load fast with quick results.
- **Reliable**
  The system should be trustworthy and give consistent outcomes for every activity analyzed.
- **Accuracy**
  The systems should be able to produce accurate results since this is directly related to children's fine motor development.
- **User Friendly**
  The system should be simple and easy to use, as non-technical users like teachers and parents will be accessing it.
- **Secured**
  Since the system will store children's information, only authorized users such as teachers and parents should be able to view the data.

4.3. Test Cases

Test Case ID	Fine Motor Level	Focus Level	Expected Output
TC 1	Need Support	Low	Child finds it hard to use hands and also cannot focus well.
TC 2	Need Support	Medium	Child needs help with hand skills but can pay some attention.
TC 3	Need Support	High	Child's hand skills are improving but focus is weak.
TC 4	Developing	Low	Child's hand skills are improving but focus is weak.
TC 5	Developing	Medium	Child is improving in both hand skills and focus.
TC 6	Developing	High	Child is improving in hand skills and has strong focus
TC 7	Advanced	Low	Child has strong hand skills but weak focus.
TC 8	Advanced	Medium	Child has strong hand skills and okay focus.
TC 9	Advanced	High	Child is good in both hand skills and focus.

Table 4:1 Test Cases

5. GNATT CHART

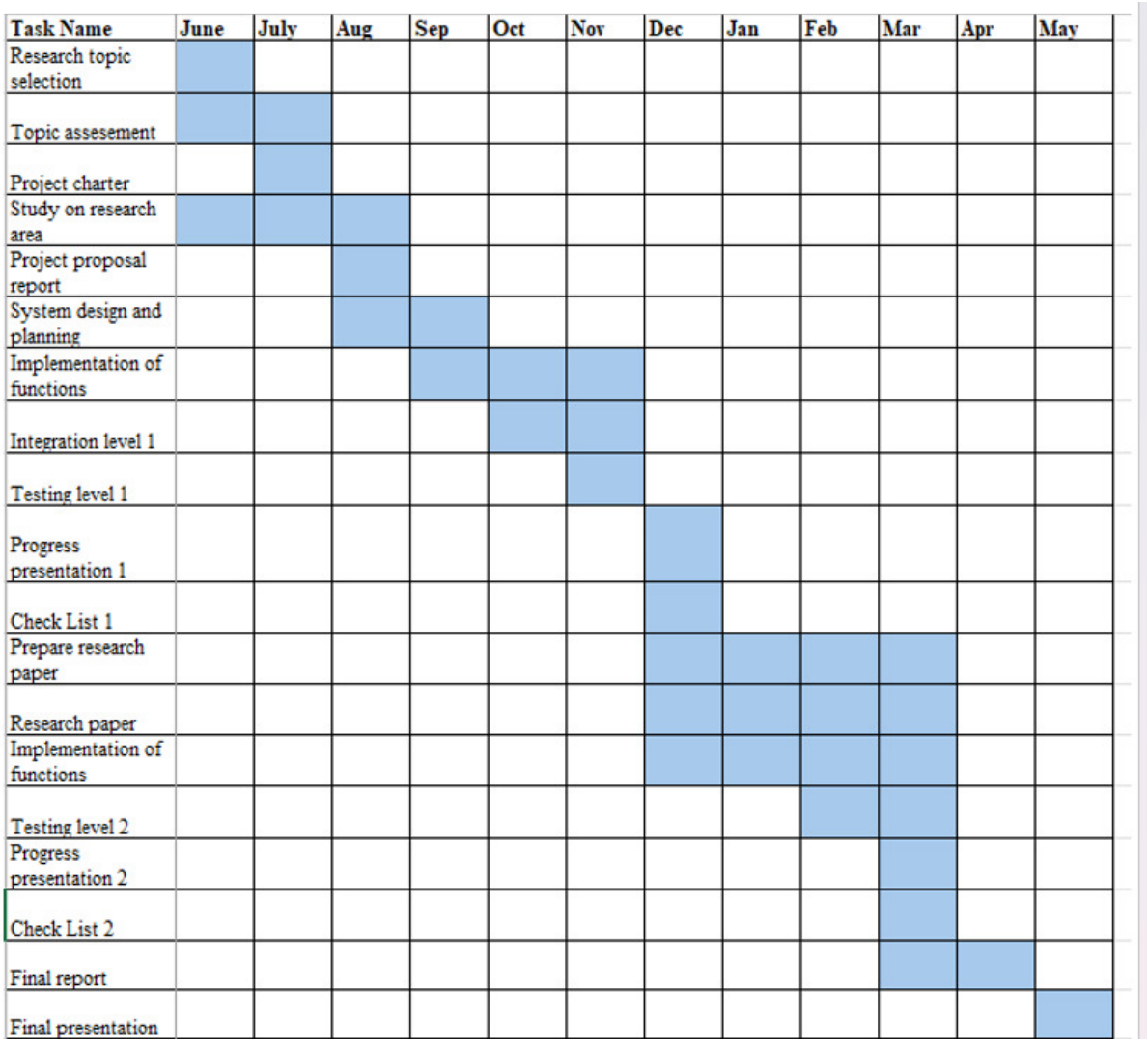


Figure 5.1 : Gnatt Chart

6. WBS

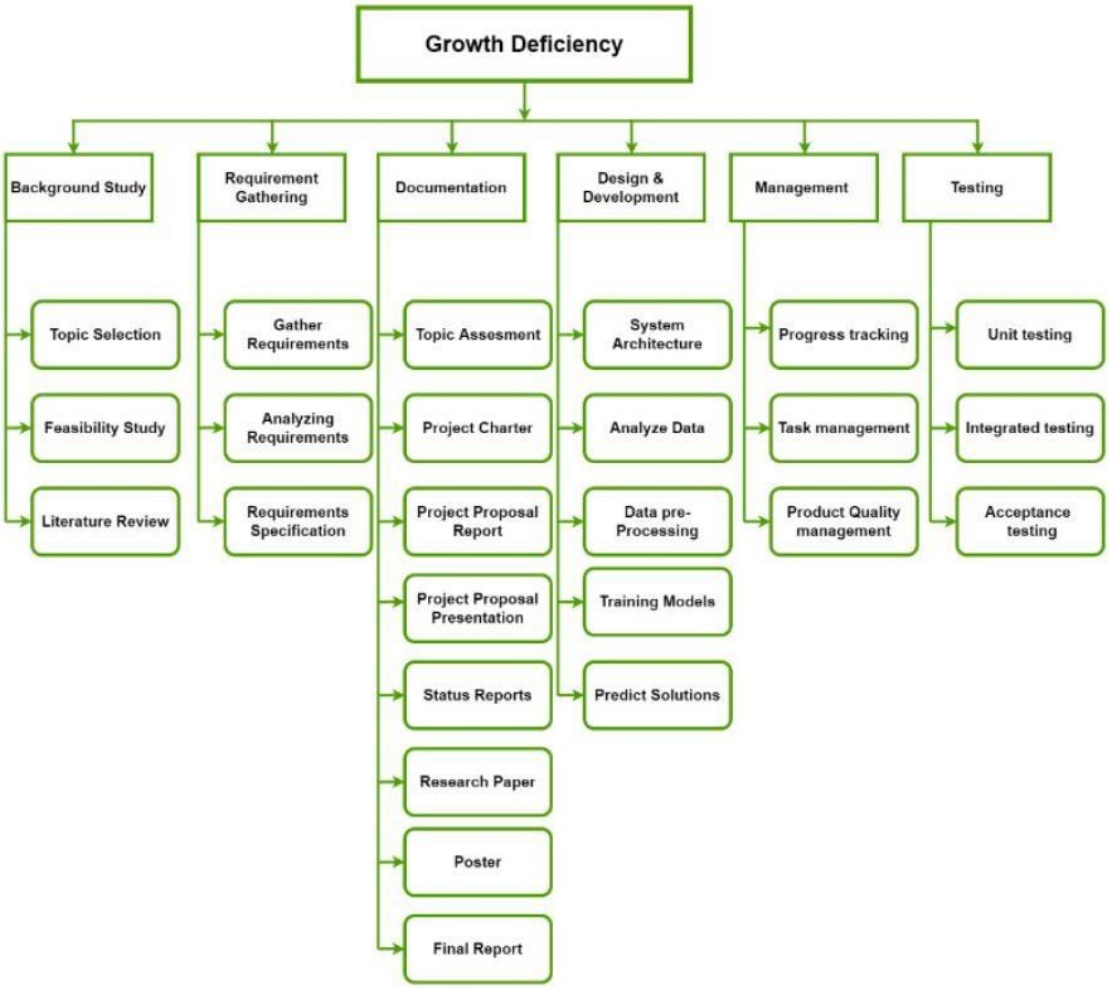


Figure 6:1 WBS

7. COMMERCIALIZATION

7.1. Customer value

Through the survey results we found out, 85% of the people would like to see an AI tool to monitor the emotional intelligence. They also highlighted how important emotional intelligence monitoring is for the early childhood development. The target audience for this application will be parents and teachers. Considering these results below are some of the steps that will be considered to commercialize the application.

7.2. Competitive analysis

Similar applications on emotional intelligence without AI models has an average price range of $9.99 monthly. Applications that use AI models has an average price of $20 monthly subscription.

How our application differ from the competitors?

Exiting applications	Our application
Gamified applications that increase digital addictions of the children	Uploading the work the child has done to reduce digital addiction
Facial expression capturing to identify emotions	Does not capture or store children's facial expressions or biometric identifiers
Use activities to improve child specific skill area.	Use activities not only to improve their skills but also to get insights on emotional intelligence based on those activities.
Limited parental involvement or communication	Facilitates and enhances parent-teacher interaction and collaboration
Focus more on cognitive, thinking, and emotions with reward-based motivation	Provides meaningful insights and personalized recommendations to parents and teachers based on the child's activity outputs
Primarily rely on digital input (typing, drag-and-drop, touch-based tasks)	Ensures that no direct digital input is involved in children's activities.
Generic design without consideration for cultural or linguistic diversity	Incorporates both Sinhala and English, ensuring cultural and linguistic appropriateness.

Table 7:1 Comparison of existing system and our application

User benefits of our applications

- Monitor child's emotional intelligence using multiple methods
- Builds emotional intelligence at an early age, which supports learning, social skills, and resilience.
- Helps teachers identify at-risk students early (e.g., children showing signs of sadness, anger, withdrawal)
- Teachers and parents both can help children to improve suggested activity skills based on dashboard feedback.

7.3. Cost analysis

On average a web hosting platform will cost $7 with postgesql database support (eg: Hostinger). This will be the rough hosting cost of our application.

7.4. Pricing plan

Considering the above factors this application will have a one month free trial period, which will be used to gain customers into the application. After the one month trial, the user will have to pay $15 monthly subscription.

Standard plan ($15) – One subscription per student.

Family plan ($30) – One subscription per three students.

School plan ($40) – Class registration ability, one subscription per 20 students.

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9. APPENDICIES

Appendix – A: MindMate Logo

Figure 9:1 Logo Design

Appendix – B: Plagiarism Result

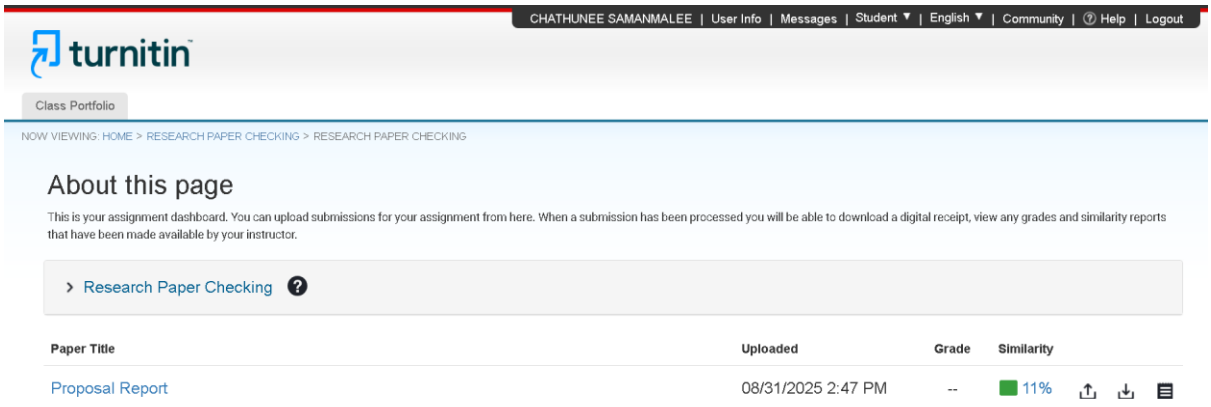

Figure 9:2 Plagiarism Result

Proposal Report

ORIGINALITY REPORT

11%	6%	3%	6%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to The University of Manchester Student Paper	3%
2	Submitted to Sri Lanka Institute of Information Technology Student Paper	1%
3	Inayah Karsa Saraswati. "Developing Childrens's Fine Motors Through Used Materials to be Useful Tools in Early Childhood", Early Childhood Research Journal (ECRJ), 2022 Publication	1%
4	scholar.archive.org Internet Source	1%
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